

Table 1. Full MGSM multilingual-transfer results for Qwen3-4B and Llama-3.2-3B-Instruct on 100 matched questions per language. Values are full-sample means $\pm$ ten-group matched-question jackknife standard errors. Higher is better for every column.

Model	Language	Method	Accuracy (%) $\uparrow$	Spanish relevance (0–2) $\uparrow$	Instruction relevance (0–2) $\uparrow$	Fluency (0–2) $\uparrow$
Qwen3-4B	Chinese	Original	71.0 $\pm$ 4.1	0.00 $\pm$ 0.00	1.80 $\pm$ 0.05	1.71 $\pm$ 0.05
		S-PID	0.0 $\pm$ 0.0	<u>2.00 $\pm$ 0.00</u>	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
		A-LQR	20.0 $\pm$ 2.6	1.98 $\pm$ 0.02	1.53 $\pm$ 0.03	1.15 $\pm$ 0.03
		H_∞ (ours)	51.0 $\pm$ 3.5	2.00 $\pm$ 0.00	1.87 $\pm$ 0.05	1.52 $\pm$ 0.04
	French	Original	60.0 $\pm$ 4.5	0.00 $\pm$ 0.00	1.93 $\pm$ 0.03	1.70 $\pm$ 0.04
		S-PID	0.0 $\pm$ 0.0	1.92 $\pm$ 0.03	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
		A-LQR	29.0 $\pm$ 5.0	1.94 $\pm$ 0.04	1.62 $\pm$ 0.05	1.11 $\pm$ 0.03
		H_∞ (ours)	48.0 $\pm$ 4.7	1.96 $\pm$ 0.04	1.96 $\pm$ 0.02	1.42 $\pm$ 0.07
	Japanese	Original	56.0 $\pm$ 4.8	0.00 $\pm$ 0.00	1.80 $\pm$ 0.04	1.64 $\pm$ 0.04
		S-PID	0.0 $\pm$ 0.0	<u>2.00 $\pm$ 0.00</u>	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
		A-LQR	32.0 $\pm$ 2.5	1.84 $\pm$ 0.06	1.63 $\pm$ 0.06	1.01 $\pm$ 0.03
		H_∞ (ours)	40.0 $\pm$ 4.9	1.98 $\pm$ 0.02	1.71 $\pm$ 0.07	1.20 $\pm$ 0.05
	Swahili	Original	15.0 $\pm$ 4.3	0.00 $\pm$ 0.00	1.43 $\pm$ 0.07	1.23 $\pm$ 0.05
		S-PID	0.0 $\pm$ 0.0	<u>1.96 $\pm$ 0.03</u>	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
		A-LQR	12.0 $\pm$ 3.6	1.40 $\pm$ 0.09	1.32 $\pm$ 0.07	0.97 $\pm$ 0.03
		H_∞ (ours)	14.0 $\pm$ 3.7	0.92 $\pm$ 0.12	1.44 $\pm$ 0.07	1.06 $\pm$ 0.04
	Telugu	Original	6.0 $\pm$ 2.7	0.00 $\pm$ 0.00	1.83 $\pm$ 0.04	1.05 $\pm$ 0.02
		S-PID	0.0 $\pm$ 0.0	<u>2.00 $\pm$ 0.00</u>	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
		A-LQR	<u>19.0 $\pm$ 2.8</u>	1.88 $\pm$ 0.04	<u>1.46 $\pm$ 0.06</u>	<u>1.18 $\pm$ 0.03</u>
		H_∞ (ours)	8.0 $\pm$ 2.5	0.34 $\pm$ 0.08	1.33 $\pm$ 0.04	1.05 $\pm$ 0.03
Llama-3.2-3B	Chinese	Original	43.0 $\pm$ 3.3	0.00 $\pm$ 0.00	1.79 $\pm$ 0.07	1.59 $\pm$ 0.07
		S-PID	9.0 $\pm$ 2.3	1.96 $\pm$ 0.03	1.23 $\pm$ 0.05	0.87 $\pm$ 0.06
		A-LQR	26.0 $\pm$ 3.7	1.80 $\pm$ 0.04	<u>1.49 $\pm$ 0.08</u>	1.21 $\pm$ 0.08
		H_∞ (ours)	22.0 $\pm$ 4.7	1.98 $\pm$ 0.02	1.46 $\pm$ 0.06	1.35 $\pm$ 0.05
	French	Original	49.0 $\pm$ 4.6	0.00 $\pm$ 0.00	1.83 $\pm$ 0.03	1.71 $\pm$ 0.04
		S-PID	14.0 $\pm$ 3.4	<u>1.98 $\pm$ 0.02</u>	1.21 $\pm$ 0.04	0.67 $\pm$ 0.06
		A-LQR	<u>43.0 $\pm$ 4.2</u>	1.24 $\pm$ 0.09	<u>1.76 $\pm$ 0.04</u>	<u>1.09 $\pm$ 0.03</u>
		H_∞ (ours)	6.0 $\pm$ 2.7	0.66 $\pm$ 0.06	0.53 $\pm$ 0.04	0.47 $\pm$ 0.04
	Japanese	Original	27.0 $\pm$ 3.0	0.00 $\pm$ 0.00	1.56 $\pm$ 0.08	1.34 $\pm$ 0.09
		S-PID	10.0 $\pm$ 2.6	1.96 $\pm$ 0.04	0.67 $\pm$ 0.06	0.63 $\pm$ 0.09
		A-LQR	<u>26.0 $\pm$ 3.7</u>	1.94 $\pm$ 0.04	<u>1.68 $\pm$ 0.08</u>	<u>1.31 $\pm$ 0.07</u>
		H_∞ (ours)	12.0 $\pm$ 3.3	1.98 $\pm$ 0.02	1.31 $\pm$ 0.10	1.16 $\pm$ 0.08
	Swahili	Original	40.0 $\pm$ 3.3	0.00 $\pm$ 0.00	1.79 $\pm$ 0.02	1.49 $\pm$ 0.03
		S-PID	2.0 $\pm$ 1.3	<u>1.90 $\pm$ 0.04</u>	0.47 $\pm$ 0.10	0.38 $\pm$ 0.05
		A-LQR	<u>22.0 $\pm$ 4.4</u>	0.68 $\pm$ 0.10	<u>1.47 $\pm$ 0.04</u>	0.98 $\pm$ 0.04
		H_∞ (ours)	14.0 $\pm$ 4.0	1.90 $\pm$ 0.04	1.28 $\pm$ 0.06	1.05 $\pm$ 0.05
	Telugu	Original	6.0 $\pm$ 2.2	0.00 $\pm$ 0.00	1.43 $\pm$ 0.04	1.07 $\pm$ 0.03
		S-PID	6.0 $\pm$ 2.7	<u>1.90 $\pm$ 0.03</u>	0.79 $\pm$ 0.08	0.46 $\pm$ 0.06
		A-LQR	<u>21.0 $\pm$ 6.0</u>	1.86 $\pm$ 0.05	<u>1.44 $\pm$ 0.08</u>	<u>1.25 $\pm$ 0.07</u>
		H_∞ (ours)	7.0 $\pm$ 2.6	0.82 $\pm$ 0.06	0.53 $\pm$ 0.10	0.54 $\pm$ 0.09